

02.4 Evidence Direction and Sample Size

Coding Protocol: Evidence Direction and Sample Size Extraction Rules

This protocol documents three record-level coding rules used across the 410 extraction records: the direction-of-finding indicator appended to each evidence quality rating; the sample size extraction convention that produces the 148-record subset used for sample-size summaries (per-record sample sizes are available in the taxonomy dataset, Appendix 03.1); and, added in version 2 (September 2026), the counting rule used to classify the 54 universal-tool records by whether their evaluated samples included moderate-risk gamblers (Section 3).

1. Evidence Direction Coding

Each extraction received an evidence quality rating (HIGH / MODERATE-HIGH / MODERATE / LOW-MODERATE / LOW) appended with a direction-of-finding indicator in parentheses. The direction indicator records the valence of the primary finding independently of the quality rating, because a well-designed study showing that a tool increases harm (negative direction) is substantively distinct from a poorly designed study showing the same result, and both are distinct from a well-designed null finding. This structure encodes negative and null findings as full members of the evidence base rather than as absences. Four direction values are used in the coding: positive, mixed, negative, and null.

1.1 Direction Value Definitions

Positive ($n = 336$; 82.0% of records)

Definition. The primary finding supports the tool's intended effect on the primary outcome or outcomes as specified by the study. This is the modal direction value and applies when the tool produced the effect it was designed to produce, under the conditions of the study.

Inclusion. Statistically significant reductions in gambling intensity or harm attributable to the tool; improvement on behavioural, psychological, or self-reported outcomes; successful implementation or deployment outcomes (e.g., uptake, compliance); positive acceptability or feasibility findings for tools evaluated on those dimensions.

Exclusion. Findings positive on one outcome but null or negative on another primary outcome ($\rightarrow$ mixed); findings positive only in a subgroup or condition where subgroup-specific effect was the main claim ($\rightarrow$ mixed); findings indicating the tool operated but did not change the outcome ($\rightarrow$ null); findings indicating harm increased or outcomes worsened ($\rightarrow$ negative).

Example. MODERATE (positive): Personalised behavioural feedback delivered via operator messaging produced significant reductions in gambling expenditure at 30-day follow-up in a large-scale operator deployment.

Mixed ($n = 53$; 12.9% of records)

Definition. The primary finding is heterogeneous across outcomes, subgroups, or implementation conditions: the tool produced positive effects on some measures and null or negative effects on others, or produced different effects in different subgroups or conditions. This is the value in which conditionally-positive effects are coded.

Inclusion. Findings positive on some primary outcomes but null on others; effects concentrated in a subgroup (e.g., high-severity but not low-severity players); effects conditional on implementation architecture (e.g., positive when default-on but null when opt-in); effects observed at one follow-up but not another; dose-dependent effects with positive effects at higher doses only.

Exclusion. Findings uniformly positive across primary outcomes ($\rightarrow$ positive); findings uniformly null ($\rightarrow$ null); findings indicating harm ($\rightarrow$ negative).

Example. LOW-MODERATE (mixed): Messages reduced erroneous beliefs at posttest but did not produce differences in actual gambling behaviour; the belief effect was concentrated in lower-severity participants.

Negative ($n = 17$; 4.1% of records)

Definition. The finding indicates the tool produced an effect opposite to the intended direction: increased rather than decreased harm indicators, worsened outcomes relative to control, or documented reactance effects. Negative findings are recorded as informative evidence, not as absence of evidence.

Inclusion. Increased gambling expenditure or session intensity associated with tool use; higher problem-gambling severity in tool-exposed groups; documented reactance or substitution effects; poorer acceptability leading to disengagement; evidence that the tool specifically worsened the primary outcome rather than simply failing to improve it.

Exclusion. Findings of no detectable effect in either direction ($\rightarrow$ null); findings where positive and negative effects co-exist across outcomes ($\rightarrow$ mixed); findings where the tool produced acceptability concerns but did not worsen gambling outcomes (code by primary outcome; acceptability concerns can be captured in the Acceptability field without changing direction).

Example. LOW-MODERATE (negative): Time-limit setting was associated with increased spending and harms at the between-person level; findings persisted after controlling for baseline severity and are consistent with reactance.

Null ($n = 4$; 1.0% of records)

Definition. The finding shows no detectable effect on the primary outcome despite methodology capable of detecting one. Null findings are distinct from negative findings (where reverse effects are detected) and from mixed findings (where effects are heterogeneous).

Inclusion. Non-significant effects on primary outcomes in studies with adequate statistical power; absence of treatment-control differences on behavioural measures; non-significant interactions in studies designed to detect them.

Exclusion. Studies too under-powered to detect a realistic effect (no direction code assigned; record may be coded LOW quality without a null finding claim); studies where some primary outcomes showed significant effects ($\rightarrow$ mixed); studies showing reverse effects ($\rightarrow$ negative).

Example. MODERATE (null): Self-exclusion reinstatement tutorial showed no significant effect on PGSI or gambling frequency versus historical control; group $\times$ time interaction $F(3,384)=0.5$, ns.

1.2 The Term “Conditional”

The term “conditional” or “conditionally positive” was used in earlier drafts of the main manuscript and appears in some working files (for example, the convergent evidence set filter was first written as “positive or conditionally positive”). It is not a fifth direction value in the coding. It is a descriptive qualifier for findings coded as mixed where the heterogeneity takes the form of “positive under specified conditions”. Mapping rule: “conditional” maps to mixed. The convergent evidence set criterion is implemented as “direction $\in$ {positive, mixed}”, which is how the main manuscript now states it.

Readers interested in the subset of mixed records where the heterogeneity is conditions-specific (rather than subgroup-specific or outcome-specific) can identify these in the record-level data by the Main_Findings and Subgroup_Effects fields.

1.3 Direction and Quality Are Coded Independently

The direction indicator is applied to all records at all quality tiers. Of the 14 MODERATE-HIGH records in the dataset, 11 are coded positive, 1 mixed, 1 negative, and 1 null: demonstrating that the

highest-quality tier in the review contains findings across all four direction values. This independence is a design feature of the protocol: direction must be coded even at lower quality tiers to distinguish weak positive evidence from weak null or negative evidence, both of which are frequently conflated in narrative reviews.

No records received the full HIGH rating in the present dataset; the highest tier populated is MODERATE-HIGH. Direction coding is unaffected by this gap because direction values are assigned independently of level. The full level $\times$ direction cross-tabulation is reported in 02.5 Evidence Quality and Readiness.

2. Sample Size Extraction Rules

The Sample_N field records the participant count for each extraction. Because responsible gambling studies span experimental designs with single-cohort participants, multi-condition experiments with split samples, longitudinal cohort studies, policy audits without participant-level data, and simulation studies with no human participants, a single numeric convention cannot accommodate all record types. The field uses a typed format convention designed to preserve information without forcing false aggregation.

2.1 Four Sample_N Format Categories

1. Single numeric N. Used when the study has one unambiguous participant count that applies to the primary analysis. The Sample_N field contains a single integer value.

Example: 363 (a single-session field study with one analytic cohort).

2. Composite N with breakdown. Used when the study reports a total along with meaningful subdivisions (condition-level, study-level, sub-sample-level) that cannot be discarded without loss of information. The Sample_N field contains a leading total followed by a parenthetical breakdown.

Example: “84 (Study 1: 46; Study 2: 38)”. Two sub-studies reported together; the leading 84 is the summed total but no single analysis used all 84 participants.

Example: “120 (60 per condition; final analytic N = 118)”. Randomisation-stage N vs analytic N; the breakdown preserves both.

3. Multi-study or multi-cohort designs without a single total. Used when the study contains multiple distinct sub-studies or cohorts and no single summed total is the meaningful unit of analysis. The Sample_N field contains the sub-totals without a leading aggregate.

Example: “Exp 1: 506; Exp 2: 1,500; Exp 3: 502”. Three independent experiments; the sum (2,508) is not a participant count because no analysis pooled the three samples.

4. Not applicable. Used when the study has no participant-level N in the conventional sense. Document analyses, policy audits, simulation-only studies, and public-health-campaign evaluations of published material fall in this category. The Sample_N field contains the phrase “Not applicable” with a brief parenthetical reason.

Example: “Not applicable (document analysis)”; “Not applicable (900 simulated players per condition)” where the simulation count is not a participant count.

Approximate values. When the study reports only an approximate total (e.g., “~200,000 gamblers”), the tilde notation is preserved in Sample_N and the record is treated under Category 3 (no single extractable integer).

2.2 Rule for Extractable N

For analyses requiring a participant count per record (e.g., median and IQR summaries, weighted aggregations), a record has an extractable N if and only if Sample_N is a single numeric value (Category 1 above). Records in Categories 2–4 are excluded from such summaries.

The rationale for this strict rule: in Category 2 (composite), the leading number is a sum that is meaningful for reporting but not necessarily for analysis. The underlying analysis may have used the sub-sample counts, the analytic N after exclusions, or a condition-specific subsample. Treating the leading number as the record's N introduces an artefactual aggregate that misrepresents the study. The same logic applies to Categories 3 and 4 with greater force.

This rule yields 148 of 410 records (36.1%) as having an extractable N. A permissive alternative (parsing the leading number from any string-formatted Sample_N with a leading numeric) would yield 375 records (91.5%) but would silently conflate aggregate, condition-level, and analytic Ns. The strict rule is more defensible for cross-record summaries but reduces coverage; the permissive alternative is available to downstream users through the record-level data.

Table 02.4.1

Extractable Sample N by Study Category

Study design	Total records	Extractable N	Not extractable	% extractable
EXP	174	44	130	25.3%
OBS	172	85	87	49.4%
QUAL	32	15	17	46.9%
POL	32	4	28	12.5%
Total	410	148	262	36.1%

Note. Extractable-N records are those with Sample_N coded as a single numeric value (Category 1). EXP records have the lowest extractable-N rate among well-represented designs because experimental designs commonly report sample sizes as condition-level splits (e.g., “60 per condition; final analytic N = 118”) which fall under Category 2. OBS and QUAL show comparable extractable-N rates (49.4% and 46.9% respectively); OBS records that are extractable typically come from operator-database studies reporting a single cohort N, while QUAL records that are not extractable typically have sub-sample composition in the parenthetical (e.g., “61 (RG experts: 22; treatment providers: 19; recovered gamblers: 20)”). POL has the lowest extractable-N rate (12.5%) because most policy audits report counts in non-participant units (e.g., sites, operators, venues) or have no participant-level N at all (Category 4); the four extractable POL records are experimental studies of policy-relevant content (e.g., harm-framing messages, demo-mode payout regulation) rather than policy audits.

2.3 Handling Multi-Sample Aggregate Statistics

For analyses that require a per-design summary of sample sizes (available per record in the taxonomy dataset, Appendix 03.1), the records with extractable N are used. Composite-N records are excluded even when a leading number is reported, for the reason given above. Aggregate summaries therefore describe the distribution of clean single-N records rather than an imputed total across all records. This choice prioritises methodological defensibility over coverage.

A downstream analyst wanting a permissive N value for all records where a leading number is present can parse the leading integer from the Sample_N string using the convention documented above. The resulting aggregate would yield different central tendency and dispersion statistics than the strict subset summary and should be clearly labelled as such to prevent comparison errors.

3. Universal-Tool Sample Severity Counting Rule

This section was added in September 2026 (version 2 of this document) in response to peer review of the main manuscript. It documents the rule used to establish, for the 54 tool records whose Tool_Target_Tier is ALL (universal tools), whether the evaluated sample included moderate-risk gamblers. The counts reported in the manuscript's Stage 1 Results (six strict, five permissive, 24 indeterminate, 19 not measured) were computed from the Data sheet of 03.1 Taxonomy Dataset using the rule below and are reproduced in Table 02.4.2; no new extraction was performed.

3.1 Denominator

The denominator is the 54 records whose tool_target_tier value begins with ALL. Each value carries a parenthetical basis for the universal classification: messaging tool (n = 25), educational tool (n = 16), general population (n = 6), structural feature (n = 5), mandatory break (n = 1), and platform program (n = 1). The 30 records coded N/A (no human participant sample) are excluded from the denominator, as are the 157 HIGH, 90 MOD, and 79 LOW records.

3.2 Classification Groups and Coding Values

The sample_severity field is free text, so each of the 54 records was assigned to exactly one of four groups by explicit rule, applied in the order listed. A record is placed in the first group whose criterion it meets.

1. Strict (coding value STRICT; n = 6; 11.1%). The sample_severity text names a moderate-risk band with a nonzero share, or restricts the sample to that band by an inclusion criterion. Example: "PGSI 3–7 required for inclusion" (the PGSI moderate-risk band); "PGSI: 17.9% moderate-high risk".

2. Permissive (coding value PERMISSIVE; n = 5; 9.3%). An instrument-based at-risk share is reported, or an at-risk threshold is applied at inclusion, without a named moderate band. Example: "SOGS; 69% no risk, 30% possible risk, 1% probable pathological"; "SGHS ≥1 required for inclusion"; "BPGS: all participants endorsed ≥2/5 items".

3. Indeterminate (coding value INDETERMINATE; n = 24; 44.4%). A severity instrument was administered, or a band-inclusive inclusion threshold was applied, but no distribution across bands is reported, so the presence of moderate-risk gamblers can be neither confirmed nor ruled out. Example: "PGSI administered; M not reported"; "PGSI <8 required for inclusion"; "PlayScan 3% high-risk, 97% lower risk" (lower risk not decomposed).

4. Not measured (coding value NOT_MEASURED; n = 19; 35.2%). Severity was not measured, not reported, or proxied by a behavioural criterion without an instrument. Example: "Not measured"; "Net loss past month required; no clinical measure".

Totals check: 6 + 5 + 24 + 19 = 54. Reading: the presence of moderate-risk gamblers in the evaluated sample is confirmed for six universal-tool records and for 11 on the permissive reading (groups 1 and 2 combined, 20.4%); it cannot be ruled out for a further 24; and for 43 of 54 records (groups 3 and 4) severity composition is not reported at the level the question requires.

3.3 Per-Record Assignments

Table 02.4.2 lists every record in the denominator with its sample_severity text as deposited in 03.1 and its assigned group. Where a paper contributes several records, each record is classified separately.

Table 02.4.2

Universal-Tool Records (Tool_Target_Tier = ALL) by Sample Severity Group

Record (study_id)	Tool code	sample_severity (as deposited in 03.1)	Group
EXP_2019_Hollingshead_FEED-BEHAV	FEED- PERSONAL + MSG-POP	PGSI 3–7 required for inclusion; M not reported	STRICT
EXP_2019_Hollingshead_MSG-REM	MSG-POP	PGSI 3–7 required for inclusion; M not reported	STRICT
EXP_2019_Hollingshead_STRUCT-PAUSE	BREAK-FORCE	PGSI 3–7 required for inclusion; M not reported	STRICT
POL_2024_Myles_EDU-INFO_DESIGN	EDU-INFO	PGSI: 17.9% moderate-high risk	STRICT
POL_2024_Myles_EDU-INFO_INDUSTRY	EDU-INFO	PGSI: 17.9% moderate-high risk	STRICT
POL_2024_Myles_EDU-INFO_NEURO	EDU-INFO	PGSI: 17.9% moderate-high risk	STRICT
EXP_2017_Broussard_EDU-INFO	EDU-INFO	SOGS; 69% no risk, 30% possible risk, 1% probable pathological	PERMISSIVE
EXP_2017_Broussard_STRUCT-DISPLAY	STRUCT-DISPLAY	SOGS; 69% no risk, 30% possible risk, 1% probable pathological	PERMISSIVE
EXP_2020_Armstrong_EDU-INTERACT_TRIAN	EDU-INTERACT	SGHS ≥ 1 required for inclusion; M not reported	PERMISSIVE
OBS_2019_Strohaker_SELF-REG	SELF-REG	Lie/Bet: 47% no indicators, 30% chasing only, 8% lying only, 15% both	PERMISSIVE
QUAL_2021_McCarthy_MSG-STATIC	MSG-STATIC	BPGS: all participants endorsed $\geq 2/5$ items	PERMISSIVE
EXP_2005_Blaszczyński_STRUCT-PAY	STRUCT-PAY	SOGS lifetime version administered; M not reported (used for classification only)	INDETERMINATE
EXP_2005_Blaszczyński_STRUCT-SPEED	STRUCT-TEMPO	SOGS lifetime version administered; M not reported (used for classification only)	INDETERMINATE
EXP_2005_Blaszczyński_STRUCT-STAKE	STRUCT-STAKE	SOGS lifetime version administered; M not reported (used for classification only)	INDETERMINATE
EXP_2012_Jardin_MSG-POP	MSG-POP	SOGS administered; M not reported	INDETERMINATE
EXP_2015_Rockloff_MSG-POP	MSG-POP	PGSI administered; M not reported	INDETERMINATE

Record (study_id)	Tool code	sample_severity (as deposited in 03.1)	Group
EXP_2019_Lavoie_MSG-POP	FEED- PERSONAL + MSG-POP	PGSI administered; M not reported	INDETERMIN ATE
EXP_2019_Lavoie_MSG-STATIC	MSG-STATIC	PGSI administered; M not reported	INDETERMIN ATE
EXP_2020_Armstrong_EDU- INTERACT_PRIME	EDU-INTERACT	PGSI administered; M not reported	INDETERMIN ATE
EXP_2020_Brown_EDU-VIDEO_AdV	EDU-VIDEO	PGSI administered; M not reported	INDETERMIN ATE
EXP_2020_Brown_EDU-VIDEO_CONTACT	EDU-VIDEO	PGSI administered; M not reported	INDETERMIN ATE
EXP_2020_Brown_EDU-VIDEO_ED	EDU-VIDEO	PGSI administered; M not reported	INDETERMIN ATE
EXP_2022_Newall_MSG-STATIC_BW	MSG-STATIC	PGSI administered; M not reported	INDETERMIN ATE
EXP_2022_Newall_MSG- STATIC_YELLOW	MSG-STATIC	PGSI administered; M not reported	INDETERMIN ATE
EXP_2023_Kiyak_MSG-POP_COG	MSG-POP	PGSI <8 required for inclusion; cognitive condition M=2.20 (SD=2.39)	INDETERMIN ATE
EXP_2023_Kiyak_MSG-POP_DIA	MSG-POP	PGSI <8 required for inclusion; cognitive condition M=2.20 (SD=2.39)	INDETERMIN ATE
EXP_2023_Kiyak_MSG-POP_INFO	MSG-POP	PGSI <8 required for inclusion; cognitive condition M=2.20 (SD=2.39)	INDETERMIN ATE
EXP_2024_Tabri_EDU-INTERACT	EDU-INTERACT + FEED-NORM	BBGS administered; M not reported	INDETERMIN ATE
OBS_2015_Gainsbury_MSG-POP_Middle	MSG-NOTIFY	PGSI administered; distribution not reported in paper	INDETERMIN ATE
OBS_2015_Gainsbury_MSG-POP_Peripheral	MSG-NOTIFY	PGSI administered; distribution not reported in paper	INDETERMIN ATE
OBS_2015_Gainsbury_MSG-POP_SA	MSG-NOTIFY	PGSI administered; M not reported	INDETERMIN ATE
OBS_2018_Auer_MSG-NOTIFY	MSG-POP + MSG-PUSH	PlayScan 3% high-risk, 97% lower risk	INDETERMIN ATE
OBS_2018_Chua_STRUCT-SPEED	STRUCT-OTHER	PGSI administered; M not reported; high-risk excluded from pilot sample	INDETERMIN ATE

Record (study_id)	Tool code	sample_severity (as deposited in 03.1)	Group
OBS_2019_Rodda_SELF-REG	SELF-REG	PGSI administered; M not reported	INDETERMINATE
OBS_2020_Delfabbro_SELF-REG-PPS	ASSESS-SELF	PGSI administered; M not reported	INDETERMINATE
EXP_2016_Auer_EDUC-RESOURCE	EDU-INFO	Net loss past month required; no clinical measure	NOT_MEASURED
EXP_2016_Auer_FEED-NORM	FEED-NORM	Net loss past month required; no clinical measure	NOT_MEASURED
EXP_2016_Auer_FEED-PERSONAL	FEED-PERSONAL	Net loss past month required; no clinical measure	NOT_MEASURED
EXP_2016_Harris_MSG-POP	MSG-POP	Not measured	NOT_MEASURED
EXP_2019_Hollingshead_EDU-VIDEO	EDU-VIDEO	Not reported	NOT_MEASURED
EXP_2024_Hopfgartner_STRUCT-SPEED	BREAK-FORCE	Not measured	NOT_MEASURED
OBS_2014_Auer_MSG-POP	MSG-POP	Not measured	NOT_MEASURED
OBS_2015_Auer_FEED-PERSONAL	FEED-PERSONAL + FEED-NORM	Not measured	NOT_MEASURED
OBS_2015_Auer_MSG-POP_Enhanced	MSG-POP	Not measured	NOT_MEASURED
OBS_2015_Auer_MSG-POP_Simple	MSG-POP	Not measured	NOT_MEASURED
OBS_2018_Turner_EDU-INTERACT	EDU-INTERACT	Not measured	NOT_MEASURED
OBS_2019_Auer_BREAK-FORCE	BREAK-FORCE	Not measured	NOT_MEASURED
OBS_2023_Auer_FEED-PERSONAL	FEED-PERSONAL	Not measured	NOT_MEASURED
OBS_2023_Louderback_PROGRAM	PROGRAM	Not measured	NOT_MEASURED
OBS_2024_Auer_MSG-NOTIFY_LW	MSG-PUSH	Not measured	NOT_MEASURED
OBS_2024_Auer_MSG-NOTIFY_WS	MSG-PUSH	Not measured	NOT_MEASURED
OBS_2024_Connolly_SELF-REG	ASSESS-SELF	Not measured	NOT_MEASURED
QUAL_2018_Gainsbury_MSG-POP	MSG-POP	Not measured	NOT_MEASURED

Record (study_id)	Tool code	sample_severity (as deposited in 03.1)	Group
			RED
QUAL_2022_Marko_MSG-STATIC	MSG-STATIC	Not measured	NOT_MEASURED

Note. Group counts: STRICT 6, PERMISSIVE 5, INDETERMINATE 24, NOT_MEASURED 19 (total 54). Groups are assigned by the rule in Section 3.2 from the sample_severity field exactly as deposited; the text is quoted verbatim so that each assignment can be checked or challenged rule by rule.